

DIGITAL SYSTEM **AND LOGIC DESIGN**

LAB ASSIGNMENT 3

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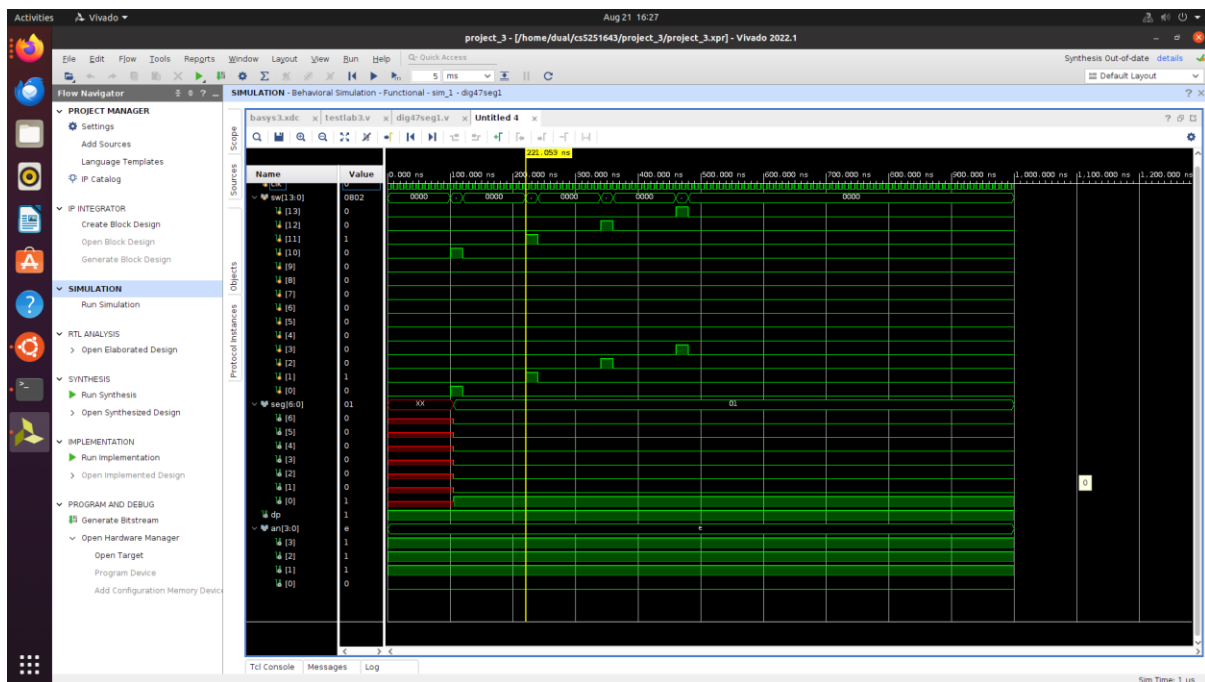
DESIGN DECISION

- USED A 14 BIT ARRAY TO STORE THE INPUTS FROM SWITCHES SW0 TO SW13
- USED A 4 BIT ARRAY TO OUPUT THE VALUE (0/1) OF EACH ANODE
- USED A 7 BIT ARRAY TO OUPUT THE VALUE (0/1) OF EACH CATHODE
- USED 4 reg VARIABLES : DATAOUT 1-4 TO STORE THE DIGIT TO BE DISPLAYED ON RESPECTIVE DECIMAL DIGIT IN 7 SEGMENT DISPLAY
- USED A 17 BIT NUMBER CALLED COUNTER TO MEASURE A CYCLE OF 1MS TO INSTANTIATE THE TURNING OFF OF CURRENT AND TURNING ON OF NEXT DISPLAY
- USED A 2BIT NUMBER ACTIVE_ANODE TO STORE WHICH OF THE 4 ANODE IS ACTIVE
- USED A 10 BIT ARRAY TO STORE WHAT IS THE DIGIT TO BE DISPLAY ON CURRENTLY ACTIVE ANODE
- RESUSED THE CODE FROM LAB2 TO SWITCH ON A PARTICULAR DIGIT OF A 7 SEGMENT DISPLAY USING A ALWAYS BLOCK
- USED A ALWAYS BLOCK TO CONSISTENTLY CHECK FOR A CHANGE IN THE DIGIT ASSOCIATED WITH ANY OF THE FOUR DIGITS BY CHECKING FOR CURRENT INPUT FOR THE ANODE(S) CURRENTLY TURNED ON .

USED A ALWAYS BLOCK WITH A CLOCK WITH A
FREQ OF 5NS TO CHECK CONSISTENTLY

- EVERY 1MS THE DECIMAL DIGIT CURRENTLY
TURNED ON SHIFTS TO THE RIGHT. THIS MEANS
EVERY DIGIT IS TURNED ON 250 TIMES EACH
SECOND WHICH IS WELL BEYOND HUMAN
FREQUENCY PERCEPTION SO WE SEE ALL FOUR
DISPLAYS SIMOLTANEOUSLY LIT
- TESTED BY SWITCHING ON DIFFERENT ANODES
WITH DIFFERENT INPUT DIGITS
- CONFIGURED BASYS3.xdc FILE TO TAKE IN INPUT
FROM SW0-13 AND OUTPUT THE 4 ANODES AND 7
CATHODES

SIMULATIONS



Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs*	32	0	0	20800	0.15
LUT as Logic	32	0	0	20800	0.15
LUT as Memory	0	0	0	9600	0.00
Slice Registers	59	0	0	41600	0.14
Register as Flip Flop	59	0	0	41600	0.14
Register as Latch	0	0	0	41600	0.00
F7 Muxes	0	0	0	16300	0.00
F8 Muxes	0	0	0	8150	0.00

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	0	0	0	50	0.00
RAMB36/FIFO*	0	0	0	50	0.00
RAMB18	0	0	0	100	0.00

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	0	0	0	90	0.00

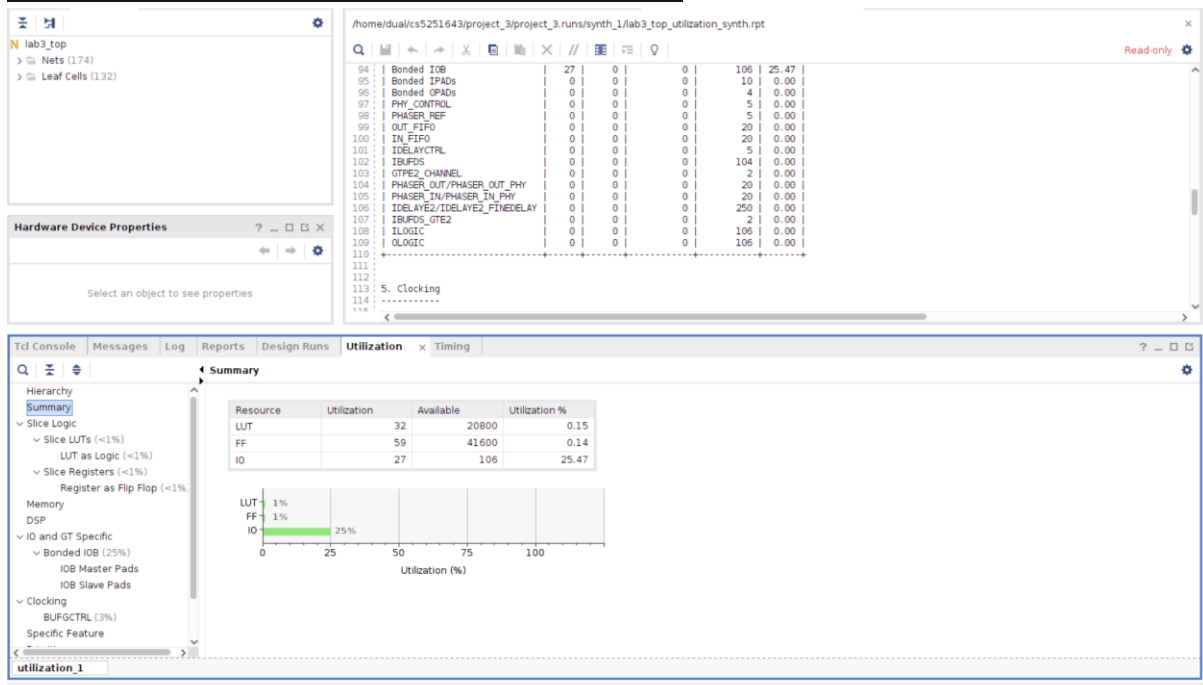
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Start RTL Component Statistics
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Detailed RTL Component Info :
+---Adders :
      2 Input   17 Bit      Adders := 1
      2 Input    2 Bit      Adders := 1
+---Registers :
              17 Bit      Registers := 1
              10 Bit      Registers := 4
              2 Bit       Registers := 1
+---Muxes :
      2 Input   17 Bit      Muxes := 1
      4 Input   10 Bit      Muxes := 1
      2 Input    7 Bit      Muxes := 1
-----
Finished RTL Component Statistics
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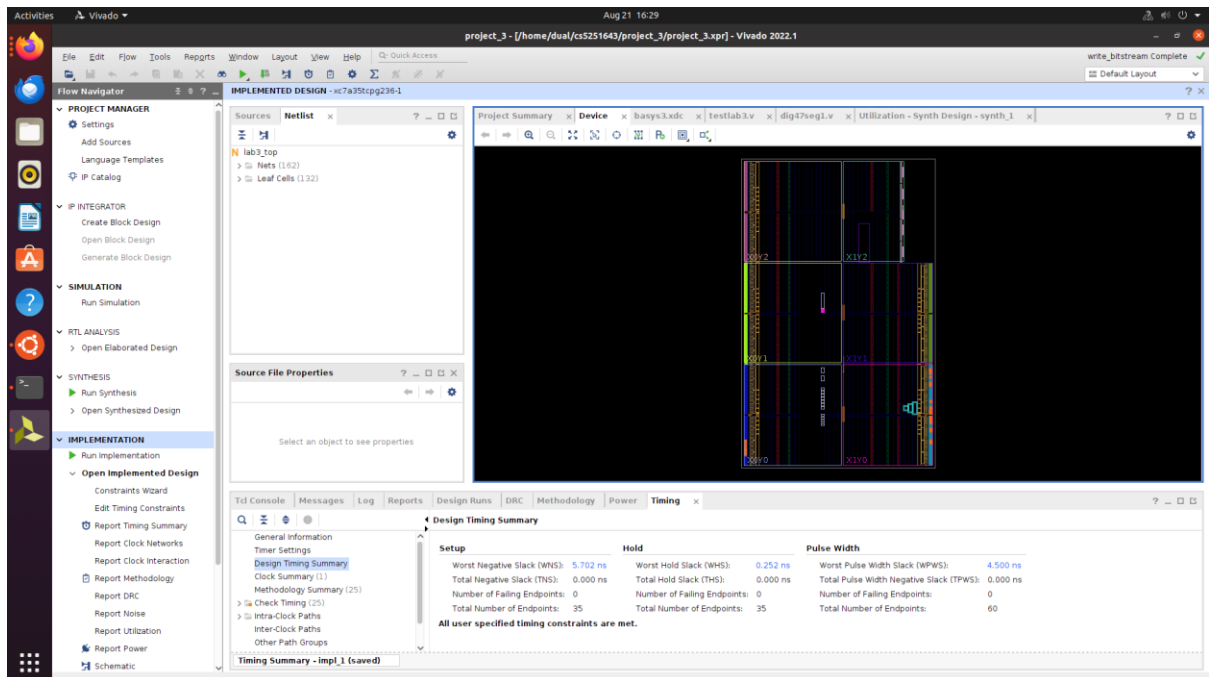
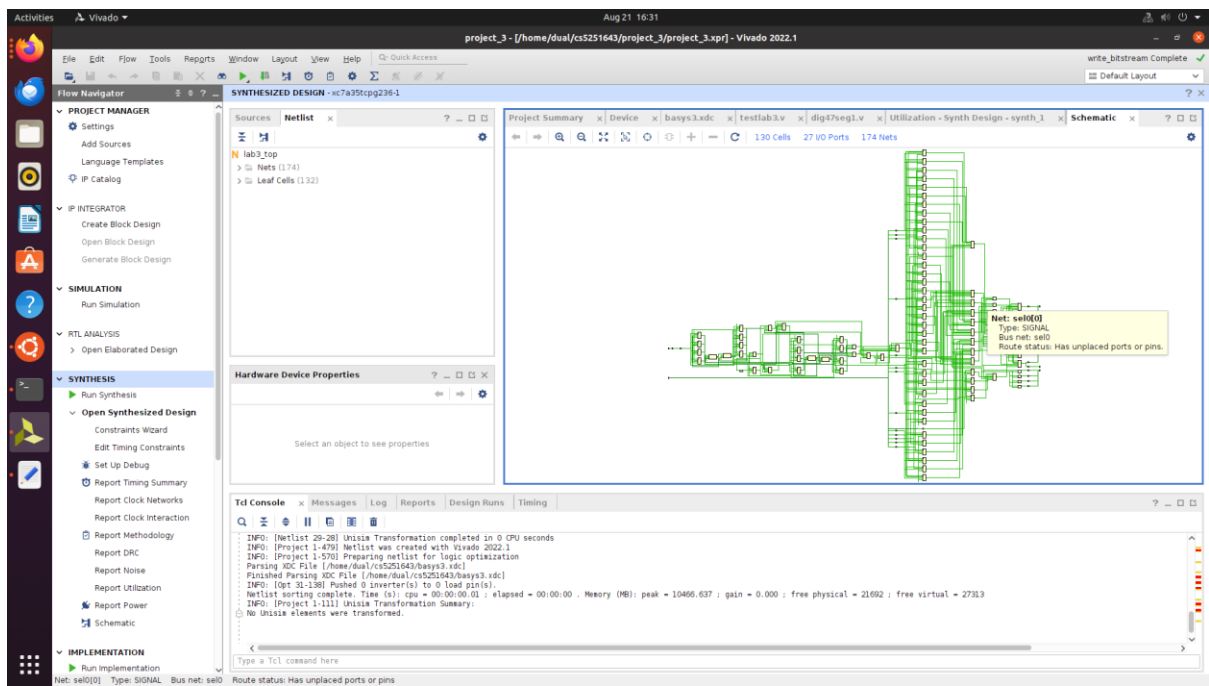
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Report Cell Usage:

	Cell	Count
1	BUFG	1
2	CARRY4	4
3	LUT1	1
4	LUT2	7
5	LUT3	6
6	LUT4	3
7	LUT5	4
8	LUT6	18
9	FDRE	59
10	IBUF	15
11	OBUF	12



GENERATED SCHEMATIC



DEBUGGING:

READ FROM TCL AND MESSAGES AND LOGS TO AND
READ THROUGH ERRORS AND CORRECTED THE CODE AT
PLACES

SAW SIMULATION AND RECTIFIED THE UNDESIRE
BEHAVIOUR OF CATHODES,ANODES,CLOCK

CONFIGURED BASYS3 PROPERLY AFTER BITSTREAM
GENERATION FAILED DUE TO IT